Author: I see that you decided to avoid referring to the concept of a variable quantity. Assume that x is a number and y is also a number. But then explain, please, the meaning of the phrase "a dependence between two numbers".

Reader: But look, the words "an independent variable" and "a dependent variable" can be found in any textbook on mathematics.

Author: The concept of a variable is given in textbooks on mathematics after the definition of a function has been introduced.

Reader: It seems I have lost my way.

Author: Actually it is not all that difficult "to construct" an image of a numerical function. I mean image, not mathematical definition which we shall discuss later.

In fact, a numerical function may be pictured as a "black box" that generates a number at the output in response to a number at the input.

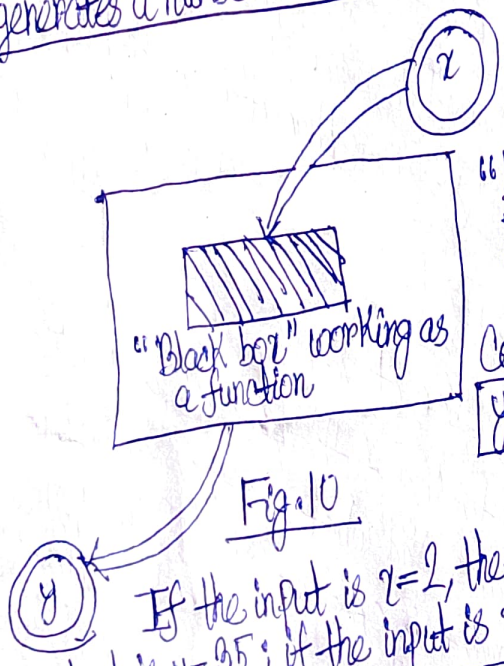

Fig. 10

You put into this "black box" a number (shown by x in Fig. 10) and the "black box" outputs a new number (y in Fig. 10).

Consider, for example, the following function: $y = 4x^2 - 1$.

If the input is $x=2$, the output is $y=15$; if the input is $x=3$, the output is $y=35$; if the input is $x=10$, the output is $y=399$, etc.

Reader: What does this "black box" look like? You have stressed that Fig. 10 is only symbolic.

Author: In this particular case it makes no difference. It does not influence the essence of the concept of a function. But a function can also be "pictured" like this: $4\square^2 - 1$